Nama : Muhammad Piyar Razaqa Aditya
Nim : 2510519031
UTS MTK SAINS DATA

Parameter Personalisasi

1. menentukan N
   $N = 31$
2. menentukan a
   $a = \text{floor}\left(\frac{31}{10}\right) = 3$
3. menentukan b
   $b = 31 \bmod 10 = 1$

4. hitung k
   $k = a + b + 1$
   $k = 3 + 1 + 1 = 5$

5. hitung θ
   $\theta_0 = (a \times 30) + (b \times 5)$
   $\theta_0 = (3 \times 30) + (1 \times 5) = 90 + 5 = 95^\circ$

6. hitung α
   $\alpha = 0.001 \times (b + 1)$
   $\alpha = 0.001 \times (1 + 1) = 0,002$

Dipindai dengan CamScanner

3

Parameter $N = 31$, $a = 3$, $b = 1$, $k = 5$

a) Himpunan

- $A = \{31, 33, 35, 38, 41, 44, 46\}$
  $B = \{34, 36, 38, 41, 43, 49\}$
  $C = \{32, 38, 41, 42, 48, 51\}$
  $U = \{31, 32, \ldots, 51\}$

a.1 $|A \cup B \cup C|$

gabungan

$\{31, 32, 33, 34, 35, 36, 38, 41, 42, 43, 44, 46, 48, 49, 51\}$

$|A \cup B \cup C| = 15$

a.2 $|A \cap B \cap C|$

irisan semua

$\{38, 41\} \rightarrow |A \cap B \cap C| = 2$

a.3 $|(A \cup B) \cap C'|$

$A \cup B = \{31, 33, 34, 35, 36, 38, 41, 43, 44, 46, 49\}$

$C = \{32, 38, 41, 42, 48, 51\}$

hilangkan elemen C nya

$= \{31, 33, 34, 35, 36, 43, 44, 46, 49\}$

a.4 $|A \oplus B|$

- $A \cap B = \{38, 41\}$
- $A \cup B = 11$ elemen

$|A \oplus B| = 11 - 2 = 9$

c) Komposisi fungsi

Fungsi = $f(x) = 5x + 4$

$g(x) = \frac{x^2}{2} + 3$

$(f \circ g)(x)$

$f(g(x)) = 5\left(\frac{x^2}{2} + 3\right) + 4 = \frac{5}{2}x^2 + 15 + 4 = \frac{5}{2}x^2 + 19$

$(g \circ f)(x)$

$g(f(x)) = \frac{(5x+4)^2}{2} + 3 = \frac{25x^2 + 40x + 16}{2} + 3 = \frac{25}{2}x^2 + 20x + 8 + 3 = \frac{25}{2}x^2 + 20x + 11$

coba $x = b + 2 = 3$:

- $f(g(3)) = \left(\frac{5}{2}\right)(9) + 19 = 22{,}5 + 19 = 41{,}5$

$g(f(3)) = \left(\frac{25}{2}\right)(9) + 20(3) + 11 = 112{,}5 + 60 + 11 = 183{,}5$

Dipindai dengan CamScanner

3

d. invers f

$$y = 5x + 4 \rightarrow x = \frac{y-4}{5}$$

$$f^{-1}(x) = \frac{(x-4)}{5}$$

$$k \cdot 100 + 4 + 6 = 500 + 4 = 504$$

$$x = \frac{504-4}{5} = \frac{500}{5} = 100$$

e. Klasifikasi Fungsi

Fungsi $f(x) = 5x + 4$
- linear dengan gradien $\neq 0$
- injektif / satu - satu
- surjektif ke R (range)

Bijektif

Fungsi $g(x) = \frac{x^2}{2+3}$

- fungsi kuadrat
- tidak injektif $(g(x) = g(-x))$
- Range $\geq 3 \rightarrow$ tidak mencakup R

Tidak surjektif ke R

bukan Bijektif